Connecting detector protection

Status and next steps

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Fermilab

Three systems, three jobs

DPS **Detector Protection System**

Stops equipment when conditions are unsafe.

SC **Slow controls**

Runs and monitors the detector.

BMS **Building Management System**

Runs and monitors building services.

Source: SC–DPS interface, v1, pp. 4–5

Detector protection needs facility information

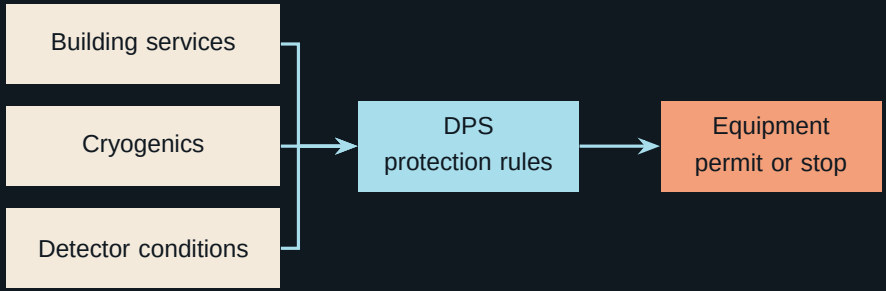


Diagram. Each protection input needs a defined meaning and owner.

Liquid level below limit; high voltage on → high voltage off.

Source: SC–DPS interface, v1, p. 13

Protection can block an operating request

Proposed control rule

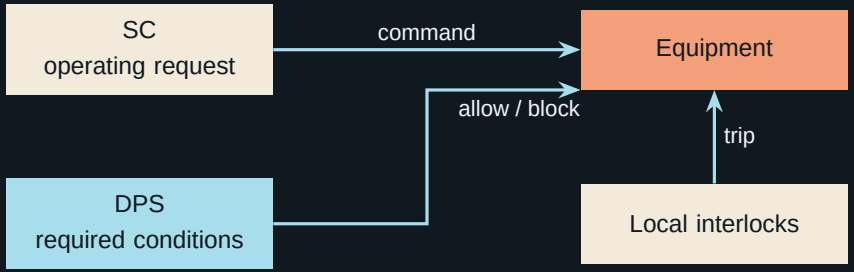


Diagram. Local trips need not wait for remote data.

SC cannot override a trip. Facility safety keeps control of its services.

Source: SC–DPS roles, v1, pp. 4–5

We have a starting point

Available	Still needed
SC–DPS interface; subsystem drafts	Agreed scope and protection rules
Facility drawings; inspection maps	Verified panels, wiring and signal lists
Hardware designs; test examples	Bench hardware and current controller projects
BMS data prototype	Live values, hosting and access

First control and interlock tables prepared. Cabinet survey pending.

Progress recorded 21 October 2026.

Source: SC–DPS interface and subsystem documents

Use early cryogenics work to agree the boundary

The early review asked for signal lists, controller locations and failure behaviour.

Use that work to define what DPS needs from cryogenics.

Agree now: each signal, action and response to lost status.

Source: Early cryogenics review charge

Source: Detector protection inputs, v1, p. 13

An interlock matrix connects a condition to an action

Condition	Action
Liquid level below limit; high voltage on	Switch off high voltage
Smoke in a rack	Cut power to that rack
Signal lost	Response to agree

Define the rule for installation, tests and operation.

Agree how fast to act, what happens on failure and who may reset.

Source: Documented examples, v1, p. 13

Each consortium reviews its protection rules

Group	Discussion
Charge readout	Cooling, power and local faults
Photon detectors	Power, temperature and lasers
High voltage	Liquid level, pressure and supply faults
Cryogenic instrumentation	Which measurements DPS needs
Data acquisition / installation	Rack protection, cooling and power

Facility teams define the building signals in both directions.

Source: Subsystem interfaces and responsibilities, v1

Build one protection path, then extend it

1

Agree

One rule, its owners and its operating stages.

2

Map

Source, controller, output and equipment feedback.

3

Test

Normal operation, faults, lost signals and recovery.

4

Extend

Verify on site, then repeat for each subsystem.

In parallel: publish agreed BMS values for experiment use.

Agree integration requirements now

Who	Next action
DPS and SC leads	Agree scope and Fermilab responsibility
Each consortium	Name a contact; review its interlock matrix
SDSTA, LBNF and cryogenics	Agree the signals and who owns them
Our team	Verify panels; test the first protection path

Agree these requirements before interfaces are fixed.

SDSTA operates SURF; LBNF is the facility project.